

# Laura H. Yang

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St. Louis, MO 63130

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<https://orcid.org/0000-0002-0057-7120>

## EDUCATION

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| <b>Harvard University</b><br><i>Ph.D.</i> , Environmental Science and Engineering | Cambridge, MA<br>09/2021 - 08/2025 |
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| <b>Georgia Institute of Technology</b><br><i>B.S.</i> , Environmental Engineering | Atlanta, GA<br>09/2016 - 12/2020 |
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## PROFESSIONAL EXPERIENCE

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| <b>Postdoctoral Research Fellow</b> , Washington University in St. Louis<br>Host: Randall V. Martin | 09/2025 - Present |
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| <b>NSF Graduate Research Fellow</b> , Harvard University<br>Advisor: Daniel J. Jacob | 09/2021 - 08/2025 |
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| <b>NSF Intern Program Research Intern</b> , NASA Jet Propulsion Laboratory<br>Hosts: Kevin W. Bowman, Kazuyuki Miyazaki | 02/2025 - 07/2025 |
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| <b>Undergraduate Research Assistant</b> , Georgia Institute of Technology<br>Advisor: Nga L. (Sally) Ng | 05/2019 - 08/2021 |
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| <b>Global Internship Program Research Assistant</b> , Yonsei University<br>Host: Jhoon Kim | 08/2019 - 08/2020 |
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## RESEARCH INTERESTS

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I am an atmospheric chemist. I use chemical transport models, satellite data, and aircraft observations to study atmospheric composition, inform air quality policy, and protect human health.

## HONORS, AWARDS, AND FELLOWSHIPS

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|-----------------------------------------------------------------------------------------------|-------------|
| Air Quality and Health Initiative Postdoctoral Fellowship, Washington University in St. Louis | 2025        |
| Intern Supplemental Funding, National Science Foundation                                      | 2025        |
| Graduate Research Fellowship, National Science Foundation                                     | 2021 - 2025 |
| Stonington Endowment Graduate Fellowship, Harvard University                                  | 2021        |
| Sigma Xi Best Undergraduate Research Award, Georgia Tech                                      | 2021        |
| Civil and Environmental Engineering Best Undergraduate Research Award, Georgia Tech           | 2021        |
| Buck Stith Outstanding Senior Award, Georgia Tech                                             | 2021        |
| President's Undergraduate Research Awards, Georgia Tech                                       | 2017, 2019  |
| Buck Stith Outstanding Junior Award, Georgia Tech                                             | 2019        |
| Callaway Scholarship, Georgia Tech                                                            | 2017        |
| Provost Scholarship, Georgia Tech                                                             | 2016 - 2020 |
| Outstanding Young Environmental Scientist Award, Cary Institute of Ecosystem Studies          | 2016        |

## PEER-REVIEWED PUBLICATIONS

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H-index: 13, citations: 455 (as of 06/2026, Google Scholar)

## First-Author Papers

- [19] **Yang, L.H.**, Jacob, D.J., Bates, K.H., Lin, H., H.M. Allen, J.F. Müller, Brown, S.S., Dang, R., Colombi N.K., Zhai, S., Yantosca, R.M., Brewer, J.F., Ng, N.L., Crounse, J.D., Wennberg, P.O., Li, K., Liao, H. Modeling of Methyl Hydroperoxide Observations in Urban and Remote Air over South Korea: Methyl Radical Chemistry and Inference of Atmospheric Methanediol. *Geophysical Research Letters*, 52(22), e2025GL118267, 2025. <https://doi.org/10.1029/2025GL118267>
- [13] **Yang, L.H.**, Jacob, D.J., Lin, H., Dang, R., Bates, K.H., East, J.D., Travis, K.R., Pendergrass, D.C., Murray, L.T. Assessment of Hydrogen's Climate Impact Is Affected by Model OH Biases. *Geophysical Research Letters*, 52(5), e2024GL112445, 2025. <https://doi.org/10.1029/2024GL112445>
- [9] **Yang, L.H.**, Jacob, D.J., Dang, R., Oak, Y.J., Lin, H., Kim, J., Zhai, S., Colombi, N.K., Pendergrass, D.C., Beaudry, E., Shah, V., Feng, X., Yantosca, R.M., Chong, H., Park, J., Lee, H., Lee, W.-J., Kim, S., Kim, E., Travis, K.R., Crawford, J.H., Liao, H. Interpreting Geostationary Environment Monitoring Spectrometer (GEMS) Observations of the Diurnal Variation in NO<sub>2</sub> over East Asia. *Atmospheric Chemistry and Physics*, 24(12), 7027–7039, 2024. <https://doi.org/10.5194/acp-24-7027-2024>
- [4] **Yang, L.H.**, Jacob, D.J., Colombi, N.K., Bates, K., Shah, V., Beaudry, E., Yantosca, R.M., Lin, H., Brewer, J., Chong, H., Travis, K., Crawford, J., Lamsal, L., Koo, J.-H., Kim, J. Tropospheric NO<sub>2</sub> Vertical Profiles over South Korea and Their Relation to Oxidant Chemistry. *Atmospheric Chemistry and Physics*, 23(4), 2465–2481, 2023. <https://doi.org/10.5194/acp-23-2465-2023>
- [3] **Yang, L.H.**, Hagan, D.H., Rivera-Rios, J.C., Kelp, M.M., Cross, E.S., Peng, Y., Kaiser, J., Williams, L.R., Croteau, P.L., Jayne, J.T., Ng, N.L. Investigating the Sources of Urban Air Pollution Using Low-Cost Air Quality Sensors at an Urban Atlanta Site. *Environmental Science & Technology*, 56(11), 7063–7073, 2022. <https://doi.org/10.1021/acs.est.1c07005>
- [2] **Yang, L.H.**, Takeuchi, M., Chen, Y., Ng, N.L. Characterization of Thermal Decomposition of Oxygenated Organic Compounds in FIGAERO-CIMS. *Aerosol Science and Technology*, 55(12), 1321–1342, 2021. <https://doi.org/10.1080/02786826.2021.1945529>
- [1] **Yang, L.H.**, Han, B.A. Data-driven Predictions and Novel Hypotheses about Zoonotic Tick Vectors from the Genus Ixodes. *BMC Ecology*, 18(7), 1–6, 2018. <https://doi.org/10.1186/s12898-018-0163-2>

## Co-Authored Papers

- [20] Lucille G.G., **Yang L.H.**, Mickley L.J., Oak Y.J. Comparing rural and urban NO<sub>2</sub> column densities from GEMS across East Asia: Diurnal and seasonal differences, *Atmospheric Environment*, 382, 122176, 2026. <https://doi.org/10.1016/j.atmosenv.2026.122176>
- [18] Dang R., Jacob D.J., Wang H., Nowlan C.R., Abad G.G., Chong H., Liu X., Shah V., **Yang L.H.**, Oak Y.J., Marais E.A., Horner R.P., Rollins A.W., Crawford J.H., Li K., and Liao H. High-resolution Geostationary Satellite Observations of Free Tropospheric NO<sub>2</sub> over North America and Implications for Lightning Emissions, *Proceedings of the National Academy of Sciences*, 122(42), e2510535122, 2025. <https://doi.org/10.1073/pnas.2510535122>
- [17] Colombi N.K., Jacob D.J., Ye X., Yantosca B., Bates K., Pendergrass D.C., **Yang L.H.**, Li K., and Liao H. Large and Increasing Stratospheric Contribution to Tropospheric Ozone over East Asia, *Atmospheric Chemistry and Physics*, 2025. <https://doi.org/10.5194/acp-26-2623-2026>
- [16] Pendergrass, D.C., Jacob, D.J., Oak, Y.J., Dang, R., **Yang, L.H.**, Beaudry, E., Colombi, N.K., Zhai, S., Kim, H., Choi, J.S., Park, J.S., Kim, S., Li, K., Liao, H. Wintertime Trends of Fine Particulate Matter

(PM<sub>2.5</sub>) in South Korea, 2021 - 2022: Response of Nitrate and Organic Components to Decreasing NO<sub>x</sub> Emissions. *Geophysical Research Letters*, 52(19), e2025GL116091, 2025.  
<https://doi.org/10.1029/2025GL116091>

[15] Westgate, S., Shivji, Z., **Yang, L.H.**, Rivera-Rios, J.C., Ng, N.L. Field evaluation of low-cost particulate matter (PM) sensor instruments in indoor and outdoor environments: A university lecture hall and urban southeastern United States *Aerosol Science and Technology*, 1–22, 2025.  
<https://doi.org/10.1080/02786826.2025.2484244>

[14] Beaudry, E., Jacob, D.J., Bates, K., Zhai, S., **Yang, L.H.**, Pendergrass, D.C., Colombi, N., Simpson, I., Wisthaler, A., Hopkins, J., Ke, L., Liao, H. Ethanol and Methanol in South Korea and China: Evidence for Large Anthropogenic Emissions Missing from Current Inventories. *ACS ES&T Air*, 2(4), 456–465, 2025.  
<https://doi.org/10.1021/acsestair.4c00210>

[12] Dang, R., Jacob, D.J., Zhai, S., **Yang, L.H.**, Pendergrass, D.C., Coheur, P., Clarisse, L., Van Damme, M., Choi, J., Park, J., Liu, Z., Xie, P., Liao, H. A Satellite-Based Indicator for Diagnosing Particulate Nitrate Sensitivity to Precursor Emissions: Application to East Asia. *Environmental Science & Technology*, 58(45), 20101–20113, 2024.  
<https://pubs.acs.org/doi/10.1021/acs.est.4c08082>

[11] Lin, H., Emmons, L.K., Lundgren, E.W., **Yang, L.H.**, Feng, X., Dang, R., Zhai, S., Tang, Y., Kelp, M.M., Colombi, N.K., Eastham, S.D., Fritz, T.M., Jacob, D.J. Intercomparison of GEOS-Chem and CAM-chem Tropospheric Oxidant Chemistry within CESM2. *Atmospheric Chemistry and Physics*, 24(15), 8607–8624, 2024.  
<https://doi.org/10.5194/egusphere-2024-470>

[10] Oak, Y.J., Jacob, D.J., Balasus, N., **Yang, L.H.**, Chong, H., Park, J., Lee, H., Lee, G.-T., Ha, E.-J., Park, R.J., Kwon, H.-A., Kim, J. A Bias-Corrected GEMS Satellite Product for NO<sub>2</sub> Using Machine Learning with TROPOMI. *Atmospheric Measurement Techniques*, 17(17), 5147–5159, 2024.  
<https://doi.org/10.5194/amt-17-5147-2024>

[8] Varon, D., Jervis, D., Pandey, S., Gallardo, S., Balasus, N., **Yang, L.H.**, Jacob, D. Quantifying NO<sub>x</sub> Point Sources with Landsat and Sentinel-2 Observations of NO<sub>2</sub> Plumes. *Proceedings of the National Academy of Science*, 121(27), e2317077121, 2024.  
<https://doi.org/10.1073/pnas.2317077121>

[7] Westervelt, D., Isavulambire, P., Yombo Phaka, R., **Yang, L.H.**, Raheja, G., Milly, G., Selenge, J.-L., Mulumba, J.P., Bousiotis, D., Djibi, B., McNeill, V.F., Ng, N.L., Pope, F., Mbela, G., Konde, J. A Low-Cost Investigation into Sources of PM<sub>2.5</sub> in Kinshasa, DRC. *ACS ES&T Air*, 1(1), 43–51, 2023.  
<https://doi.org/10.1021/acsestair.3c00024>

[6] Zhai, S., Jacob, D.J., Pendergrass, D., Colombi, N., Shah, V., **Yang, L.H.**, Zhang, Q., Sun, Y., Wang, S., Luo, G., Yu, F., Woo, J.-H., Kim, Y., Kim, H., Dibb, J.E., Lee, T., Han, J.-S., Li, K., Liao, H. Coarse Particulate Matter Air Quality in East Asia: Implications for Fine Particulate Nitrate. *Atmospheric Chemistry and Physics*, 23(7), 4271–4281, 2023.  
<https://doi.org/10.5194/acp-23-4271-2023>

[5] Colombi, N.K., Jacob, D.J., **Yang, L.H.**, Zhai, S., Shah, V., Grange, S.K., Yantosca, R.M., Kim, S., Liao, H. Why Is Ozone in South Korea and the Seoul Metropolitan Area So High and Increasing? *Atmospheric Chemistry and Physics*, 23(7), 4031–4044, 2023.  
<https://doi.org/10.5194/acp-23-4031-2023>

## REPORT

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**Yang, L.H.**, Kim J, Ahn D.H., 2021. Air pollution: an introduction to its causes, effects, and solutions. Ministry of Foreign Affairs of the Republic of Korea, National Council on Climate and Air Quality of the Republic of Korea. Government Publications Registration Number: 11-1262000-000285-01.

## PRESENTATIONS

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- [20] Evolving Sensitivity of PM<sub>2.5</sub> to Emission Controls in a Changing Atmospheric Chemical Regime. The 12th International GEOS-Chem Meeting, St. Louis, MO, June 2026. (Talk)
- [19] Assessment of Hydrogen's Climate Impact Is Affected by Model OH Biases. Methane Emissions Technology Alliance (META) led by Stanford University and Lawrence Berkeley National Laboratory, Remote, February 2026. (Invited Talk)
- [18] Megacities as Indicators of National NO<sub>x</sub> and CO<sub>2</sub> Emission Trends. AGU Annual Meeting, New Orleans, LA, USA, December 2025. (Talk)
- [17] Advancing Chemical Transport Modeling for Air Quality, Satellite Retrievals, and the Clean Energy Transition. Center for Aerosol and Engineering (CASE) seminar, St. Louis, MO, USA, November 2025. (Talk)
- [16] Modeling of Methyl Hydroperoxide Observations in Urban and Remote Air over South Korea: Methylperoxy Radical Chemistry and Inference of Atmospheric Methanediol. Caltech Laboratory for the Study of Atmospheric Composition Group Meeting, Pasadena, CA, USA, June 2025. (Invited Talk)
- [15] Co-evolution of Fossil Fuel CO<sub>2</sub> and NO<sub>x</sub> in Megacities around the World. NASA JPL Earth Action for Air Quality Meeting, La Cañada Flintridge, CA, USA, May 2025. (Talk)
- [14] Advancing Chemical Transport Modeling for Air Quality, Satellite Retrievals, and the Clean Energy Transition. NASA JPL Tropospheric Composition Group Meeting, La Cañada Flintridge, CA, USA, May 2025. (Talk)
- [13] Assessment of Hydrogen's Climate Impact Is Affected by Model OH Biases. EGU Annual Meeting, Vienna, Austria, April 2025. (Talk)
- [12] Modeling of methyl hydroperoxide observations in urban and remote air over South Korea: Demonstration of high-NO and low-NO regimes for hydrocarbon oxidation, and inference of atmospheric methanediol. AMS Annual Meeting, New Orleans, LA, USA, January 2025. (Talk)
- [11] Assessment of Hydrogen's Climate Impact Is Affected by Model OH Biases. AGU Annual Meeting, Washington, DC, USA, December 2024. (Talk)
- [10] Interpreting GEMS Geostationary Satellite Observations of the Diurnal Variation of Nitrogen Dioxide (NO<sub>2</sub>) over East Asia. The 104th AMS Annual Meeting, Baltimore, MD, USA, January 2024. (Poster)
- [9] Interpreting GEMS Geostationary Satellite Observations of the Diurnal Variation of Nitrogen Dioxide (NO<sub>2</sub>) over East Asia. The 14th GEMS Science Team Meeting, Jeju, South Korea, September 2023. (Talk)
- [8] Interpreting GEMS Geostationary Satellite Observations of the Diurnal Variation of Nitrogen Dioxide (NO<sub>2</sub>) over East Asia. NASA and EPA Joint Atmospheric Chemistry Group Meeting, Remote, December 2023. (Invited Talk)
- [7] Tropospheric NO<sub>2</sub> Vertical Profiles over South Korea and Their Relation to Oxidant Chemistry: Implications for Geostationary Satellite Retrievals and the Observation of NO<sub>2</sub> Diurnal Variation from Space. Joint Science Meeting for TEMPO, GeoXO, & TOLNet, Huntsville, AL, USA, May 2023. (Poster)
- [6] Tropospheric NO<sub>2</sub> Vertical Profiles over South Korea and Their Relation to Oxidant Chemistry: Implications for Geostationary Satellite Retrievals and the Observation of NO<sub>2</sub> Diurnal Variation from Space. Atmospheric Measurement Group Meeting, Center for Astrophysics – Harvard & Smithsonian, Cambridge, MA, USA, December 2023. (Invited Talk)
- [5] Tropospheric NO<sub>2</sub> Vertical Profiles over South Korea and Their Relation to Oxidant Chemistry: Implications for Geostationary Satellite Retrievals and the Observation of NO<sub>2</sub> Diurnal Variation from Space. AGU Fall Meeting, Chicago, IL, USA, December 2022. (Poster)
- [4] Tropospheric NO<sub>2</sub> Vertical Profiles over South Korea and Their Relation to Oxidant Chemistry: Implications for Geostationary Satellite Retrievals and the Observation of NO<sub>2</sub> Diurnal Variation from Space. The 13th GEMS Science Team Meeting, Seoul, South Korea, November 2022. (Talk)
- [3] Tropospheric NO<sub>2</sub> Vertical Profiles over South Korea and Their Relation to Oxidant Chemistry: Implications for Geostationary Satellite Retrievals and the Observation of NO<sub>2</sub> Diurnal Variation from

Space. The 10th International GEOS-Chem Meeting, St. Louis, MO, June 2022. (Talk)

[2] Understanding the Sources of Urban Air Quality Using Low-Cost Air Quality Sensors. The 39th AAAR Conference, Remote, October 2021. (Talk)

[1] Thermal Decomposition Characterization of Filter Inlet for Gases and AEROSols (FIGAERO) coupled with Chemical Ionization Time-of-Flight Mass Spectrometer (ToF-CIMS). The 37th AAAR Conference, Portland, OR, October 2019. (Poster)

## MENTORSHIP

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**Jaden Southern** (Stanford University, class of 2026) Summer 2024

Co-mentor with Ruijun Dang, Yujin Oak, and Loretta Mickley.

Project: Evaluating TEMPO Satellite HCHO and NO<sub>2</sub> Products Against Ground-Based Pandora Instruments.

**Lucy Gagnon** (Williams College, Class of 2024; now Ph.D. student at Duke University) Summer 2023

Co-mentor with Yujin Oak and Loretta Mickley.

Project: Spatial and Temporal Differences in NO<sub>2</sub> Column Densities and Implications for Geostationary Satellite Product Applications Across Asia.

## TEACHING EXPERIENCE

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**Harvard University**, Teaching Fellow

Cambridge, MA

*EPS 200: Atmospheric Chemistry and Physics*

Spring 2023, Spring 2024

**Georgia Institute of Technology**, Teaching Assistant

Atlanta, GA

*CEE 2300: Environmental Engineering Principles*

Spring 2018, Spring 2019

**Georgia Institute of Technology**, Grader

Atlanta, GA

*ME 3322: Thermodynamics*

Spring 2019

## SERVICE

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**Reviewer** *Atmospheric Measurement Techniques; npj Clean Air; Environmental Science: Atmospheres*  
NASA review panel (2025)

**Leader** *Group Life and Community Subgroup*, Harvard Atmospheric Chemistry Modeling Group (ACMG) (2024 - 2025)

**Organizer** *Journal Club*, Harvard ACMG (2022–2024)  
*Graduate Student Forum*, Harvard ACMG (2022–2024)  
*Undergraduate Research Symposium 2024*, Harvard ACMG (2024)

**Member** American Geophysical Union, American Meteorological Society, Sigma Xi

**Volunteer** *Science Club for Girls*, Boston (2021)